



XperRAM Quick Manual

for NanoSpectrum v2.6 and earlier



XperRAM 사용자 가이드

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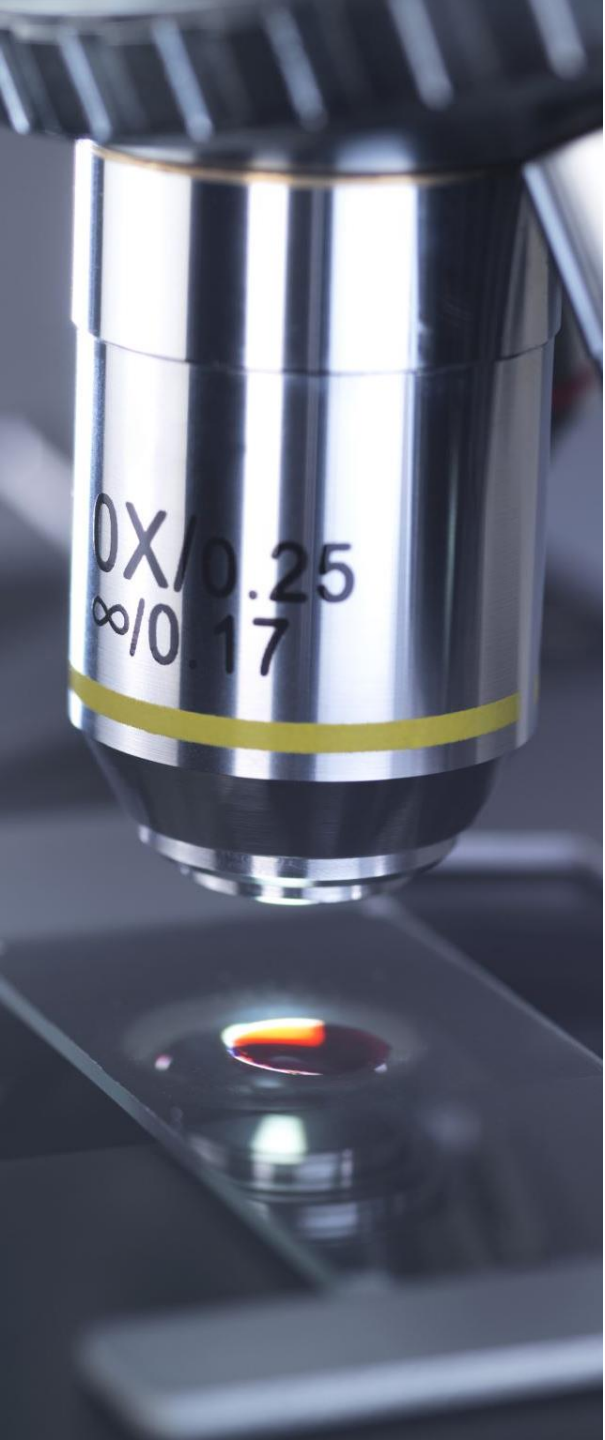
Page 9. CCD Binning 조작 방법

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Quick manual for the XperRAM system check

1. Turn on the system (Microscope, laser controller, and scanning controller)
2. Set *the laser power to 2 mW* (at the sample side) using a power meter.
3. Run the 'NanoSpectrum' software on the computer system.
4. Place *a silicon reference sample* on the microscope stage.
5. Set the *exposure time as 500 ms*.
6. Set the microscope filter to #1 and the mode lever all the way in.
7. Focus on the sample surface *using a 40X objective lens*.
8. Set the microscope filter to #1 and the mode lever halfway out.
9. Make the good laser focus on the sample surface.
10. Set the microscope filter to #2 and the mode lever all the way out.
11. *Acquire a Raman spectrum* by clicking on the 'Take Spectrum' icon.

If you want to have additional checks for mapping performance.

12. Set a mapping area of *50 μm x 50 μm with a step distance of 1 μm* .
13. Set channel conditions for the peak of the silicon (520 cm^{-1})
14. *Take a mapping image* by clicking on the Mapping icon.

Frequently Asked Question #1

Can not get the optical image of the sample

Possible cause	Solution
<i>'Vision mode' is not set</i>	<i>Set the microscope filter to #1 and the mode lever all the way in</i>
<i>The LED lamp is not turned on</i>	<i>Switch the microscope on</i>
<i>The LED source is physically shuttered</i>	<i>Open microscope shutter</i>
<i>The vision camera is not connected</i>	<i>Aming tap → General → Select 'Daheng630'</i>
<i>The vision camera is in capture mode</i>	<i>Vision window → Click on the camera capture icon to initiate camera imaging</i>
<i>Poor cable connection</i>	<i>Check if the USB3 vision digital camera is recognized in Device Manager.</i>
<i>Not enough light intensity</i>	<i>Increase the light intensity by adjusting the LED brightness using the knob on the microscope.</i>

Frequently Asked Question #2

Can not see the focused laser beam spot

<i>Possible cause</i>	<i>Solution</i>
<i>Laser is off</i>	<i>Check the electrical power connection of the laser</i>
<i>Laser output power is too low</i>	<i>Increase the output laser power using the dial switch</i>
<i>The laser driver is off</i>	<i>Turn on the laser driver and switch on the key</i>
<i>The laser is blocked by the ND filter</i>	<i>Readjust the manual ND filter</i>
<i>'Vision + Raman mode' is not set</i>	<i>Set the microscope filter to #1 and the mode lever to the middle position.</i>
<i>Misuse of non-corresponding filter box.</i>	<i>Replace with a filter box matching the laser wavelength. (XperRAM-S, -RF)</i>
<i>Misaligned beam</i>	<i>Beamline realignment is required (contact Nanobase)</i>

Frequently Asked Question #3

Raman signal intensity is too low

Possible cause	Solution
<i>The CCD is not connected</i>	<i>Check the CCD connection in a Nanospectrum and set the CCD as Andor (for XperRAM-S, -RF) or Artemis (for XperRAM-C)</i>
<i>The CCD binning condition is not good</i>	<i>Acquisition → CCD binning → adjusts the ROI area to the signal spot</i>
<i>Misuse of non-corresponding filter box.</i>	<i>Replace with a filter box matching the laser wavelength. (XperRAM-S, -RF)</i>
<i>The RF conversion switch is up.</i>	<i>Push the switch down in the RF switching box. (XperRAM-RF)</i>
<i>The grating is not inserted</i>	<i>Insert grating and adjust a right grating angle. (refer to the appendix #1)</i>
<i>The strong fluorescence signal</i>	<i>Try to measure with a longer excitation wavelength or Use another analytical technique</i>
<i>The sample is Raman inactive</i>	<i>Use another analytical technique</i>
<i>The mechanical slit is too much closed</i>	<i>Readjust the mechanical slit to have optimal signal quality.</i>

Frequently Asked Question #4

Laser spot does not move in X-Y direction

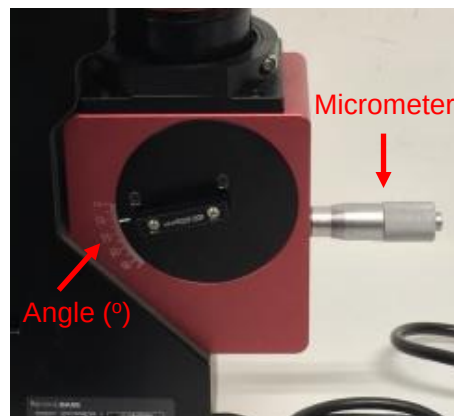
Possible cause	Solution
<i>The scanner controller power is off</i>	<i>Turn on the scanner controller</i>
<i>The scanner controllers not recognized</i>	<i>Go to Aiming → Setting → “Select ScannerAxis” for both X and Y axis</i>
<i>Original coordinates lost</i>	<i>Go to Aiming → Setting → Motion axis → Load defaults</i>
<i>Communication error</i>	<i>Inspect the communication cable between the scanner controller and the scanner body</i>



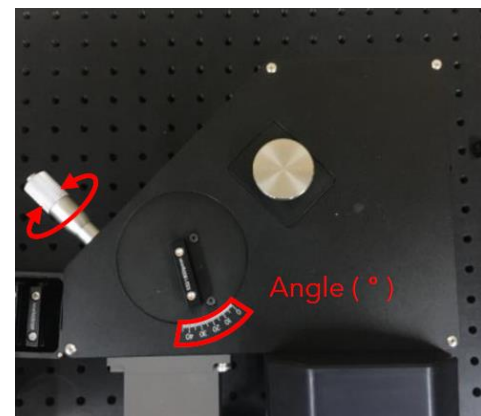
How to exchange grating

1. Confirm the spectral range and resolution of the grating that you want to use. (refer to the appendix #1)
2. Remove (pull out) the currently inserted grating from the spectrometer.
3. Insert (push in) the new grating into the slot in the spectrometer.
4. Confirm the grating angle to be set. (refer to the appendix #1)
5. Adjust a grating angle using a micrometer knob on the spectrometer.
6. Perform an "X-calibration"

XperRAM-C



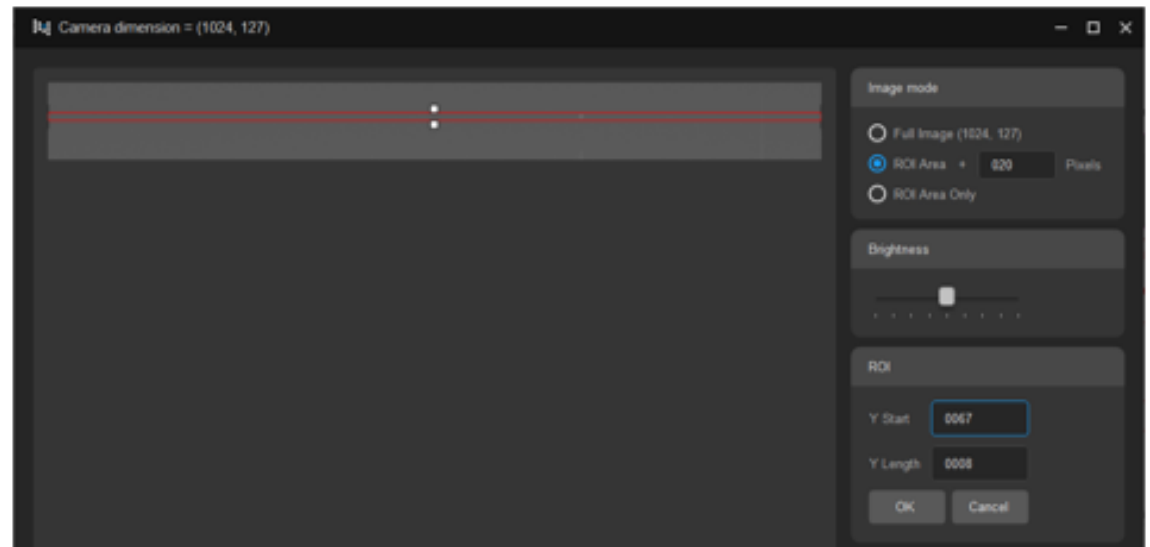
XperRAM-S, -RF



How to do CCD binning

1. Perform the "XperRAM condition check" and then proceed with the following steps.
2. Click on 'Acquisition' in the top menu bar, then select 'CCD Binning'.
3. Change to 'Full Image' in the image mode.
4. Increase 'Brightness' until the signal spots are visible on the detector display.
5. Modify the value of 'Y start' in the ROI to move the red-colored ROI area
6. Adjust the ROI area to fully cover the signal spot on a detector display.

Caution. The recommended value for the height of the ROI area 'Y length' is 8 to 10.





How to do X-Calibration

This is a brief guide to demonstrate how to perform wavenumber calibration (X-calibration) using reference samples of 'Acetaminophen' and 'silicon' after changing the grating or laser.

1. *Insert a new grating that you want to use into the slot in the spectrometer.*
2. *Manually adjust the grating angle (refer to the appendix #1)*
3. *Perform the "XperRAM condition check".*
4. *Perform the "CCD binning"*
5. *Acquire a spectrum of acetaminophen (use the reference sample)*
6. *Set the number of peaks for calibration. (Using 25 peaks is recommended)*
7. *Click on 'Calibration' the top menu bar, then select 'X-Calibration'*
8. *Select a correct reference file in the Reference File List.*
9. *Align the direction of the measured spectrum to be matched with the reference.*
10. *Enter the values for error (recommend value: 0.1) and laser wavelength.*
11. *For the automatic calibration, click the 'Auto' button.*
(For the manual calibration, please refer to the manual book.)
12. *Wait until the message "calibration success" appears.*
(This process may take 1~2 min to complete, depending on the PC performance)
13. *Measure the silicon sample and verify the peak position of silicon at 520 cm⁻¹*



How to do Scanner Calibration

Ground vibrations and repeated temperature changes from the outside can impact the system's performance. Here's a quick guide on performing scanner calibration to ensure the position accuracy calibration of the focused laser beam.

1. Power on the system (Microscope, laser controller, and scanning controller)
2. Place a [silicon reference sample](#) on the microscope stage.
3. Set the microscope filter to #1 and the mode lever to the middle position.
4. Ensure the laser is well-focused on the silicon surface.
5. [Lower the laser power](#) until the size of the laser spot is minimized using and ND filter
6. [Go to Aiming → Setting → Motion axis](#)
7. Click on the gear icon corresponding to the X-axis.
8. [Choose "Run X Calibration."](#)
9. Click "Apply" once the automatic motion calibration is finished.
10. [Repeat steps 7 to 9 for the Y-axis.](#)
11. Scanner calibration is now complete.

How to use optical modulators

XperRAM-C



For the input laser modulation, there are three available slots. (two slots in a system body are for ND-filter and polarizer and an additional slot in a microscope is only for the waveplates)

For the output signal modulation, there is a slot available in the system body. (A slot in a system body)

The optical modulators

- ND filter
- Half and Quarter waveplate
- Polarizer
- Etc.

Optical modulator accessories

Angle adjustable guide frame



Appendix #1

System	Spectrometer	Laser (nm)	Grating (lpmm)	Raman shift			PL spectrum			Grating angle
				Spectral range (cm ⁻¹)	Spectral resolution (cm ⁻¹)		Spectral range (nm)	Spectral resolution (nm)		Angle (°)
					FWHM @center	per pixel @center		FWHM @center	per pixel @center	
XperRAM-S	XPE200	405	2400	2,884	5.6	1.4	458	0.104	0.027	31.2
			1800	3,925	7.6	1.9	481	0.148	0.038	23.5
			1200	5,653	10.8	2.8	525	0.233	0.060	18.2
			600	9,365	17.2	4.4	651	0.478	0.123	9.1
			300	13,635	22.6	5.8	901	0.964	0.248	5.6
		532	2400	1,519	2.9	0.8	579	0.090	0.023	41.7
			1800	2,237	4.3	1.1	604	0.139	0.036	30.7
			1200	3,395	6.5	1.7	649	0.227	0.058	22.6
			600	5,945	11.1	2.9	777	0.475	0.122	11.3
			300	8,430	15.9	4.1	961	0.962	0.248	6.7
		633	1800	1,512	2.9	0.8	700	0.129	0.033	36.8
			1200	2,416	4.7	1.2	747	0.221	0.057	26.1
			600	4,404	8.3	2.1	876	0.472	0.121	13.1
			300	5,431	12.4	3.2	962	0.961	0.247	7.6
		785	1200	1,549	3.0	0.8	893	0.210	0.054	30.1
			600	2,372	5.7	1.5	963	0.466	0.120	15.7
XperRAM C	XPE35	405	2400	4392	5.6	2.25	800	0.112	0.045	32.6
			1800	5853	7.4	2.97	800	0.161	0.065	24.9
			1200	8121	10.0	4.04	800	0.254	0.102	21.1
			600	12420	14.1	5.70	800	0.523	0.211	10.5
		532	2400	2348	3.0	1.21	800	0.097	0.039	43.2
			1800	3404	4.3	1.74	800	0.150	0.061	32.2
			1200	5006	6.3	2.53	800	0.246	0.099	25.5
			600	8150	9.7	3.90	800	0.519	0.209	12.7
		633	1800	2325	3.0	1.20	800	0.139	0.056	38.3